



សាកលវិទ្យាល័យអ៊ីនធឺណេស្យុងនៃភ្នំពេញ
PHNOM PENH INTERNATIONAL UNIVERSITY

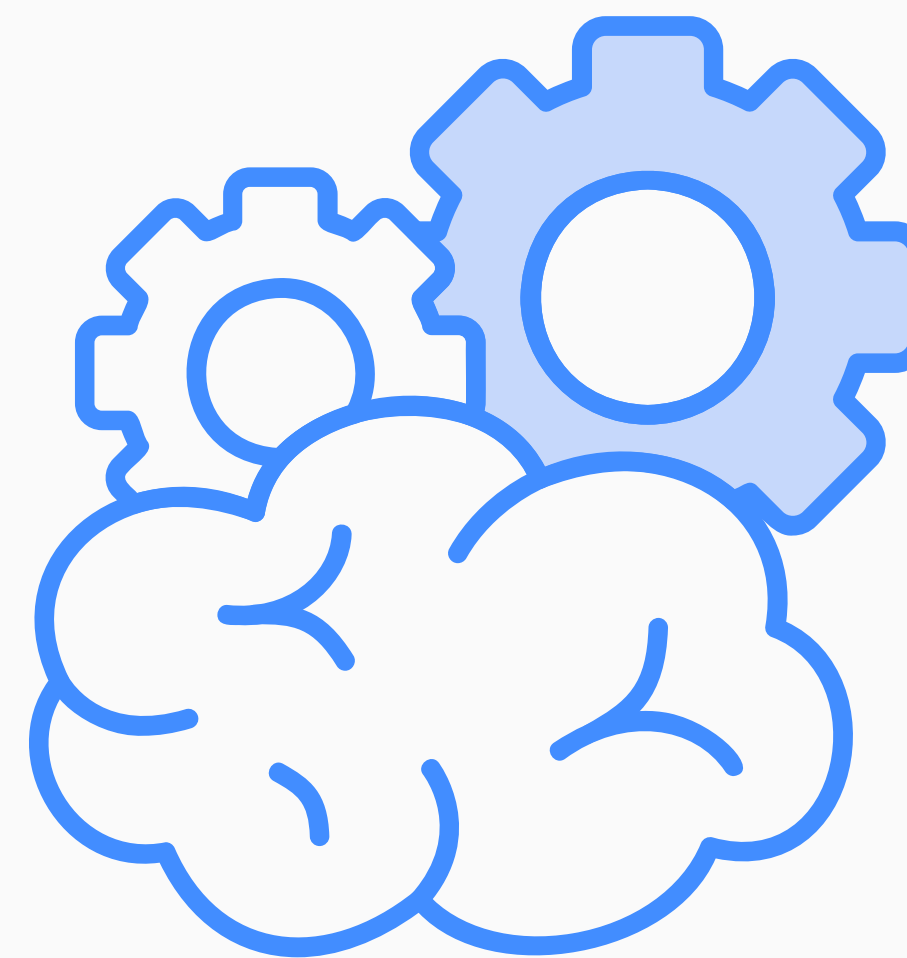
DIABETES PREDICTION

Class : Machine Learning

Lecturer : Him Soklong

Presented By Group 2

- Ly Seavmey
- Chamroeun Vichita



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1.INTRODUCTION

- Diabetes is one of the most common chronic diseases worldwide. Early prediction can help doctors diagnose patients earlier and improve treatment outcomes.
- The objective of this project is to build machine learning models that predict whether a patient has diabetes based on medical information and compare the performance of Logistic Regression and Random Forest.
- Problem Type: **Classification**

1.INTRODUCTION

- **Dataset Description:**
- **Source:** Kaggle
- **Dataset:** Diabetes Prediction Dataset

Target Variable (CLASS):

0 = Non-Diabetic

1 = Prediabetic

2 = Diabetic

- **Key Features:** HbA1c, BMI, AGE, Glucose, Urea, Cr (Creatinine), Chol, TG, HDL, LDL, VLDL, Gender.

2.DATA INSIGHTS

Features

Column	Data type	Description
ID	int64	Unique patient identifier
No_Pation	int64	Patient number/record ID
Gender	object	Male or Female
AGE	int64	Patient's age
Urea	float64	Blood urea level

2.DATA INSIGHTS

Features

Column	Data type	Description
Cr	int64	Creatinine level
HbA1c	float64	Average blood glucose level over the past 2–3 months
Chol	float64	Cholesterol level
TG	float64	Triglyceride level
HDL	float64	High-density lipoprotein ("good" cholesterol)

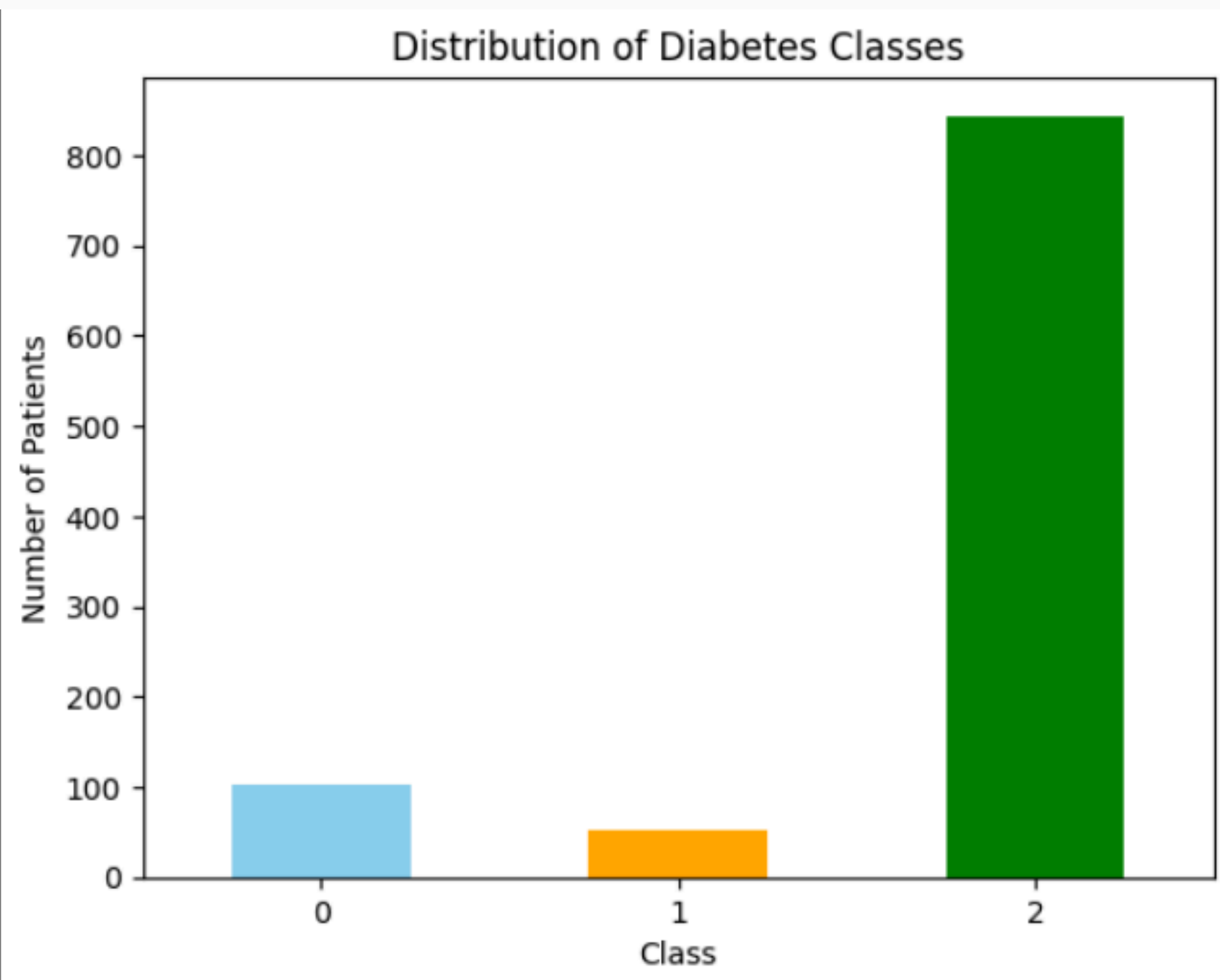
2.DATA INSIGHTS

Features

Column	Data type	Description
LDL	float64	Low-density lipoprotein ("bad" cholesterol)
VLDL	float64	Very low-density lipoprotein
BMI	float64	Body Mass Index
CLASS	object	Target variable (Diabetic status)

2.DATA INSIGHTS

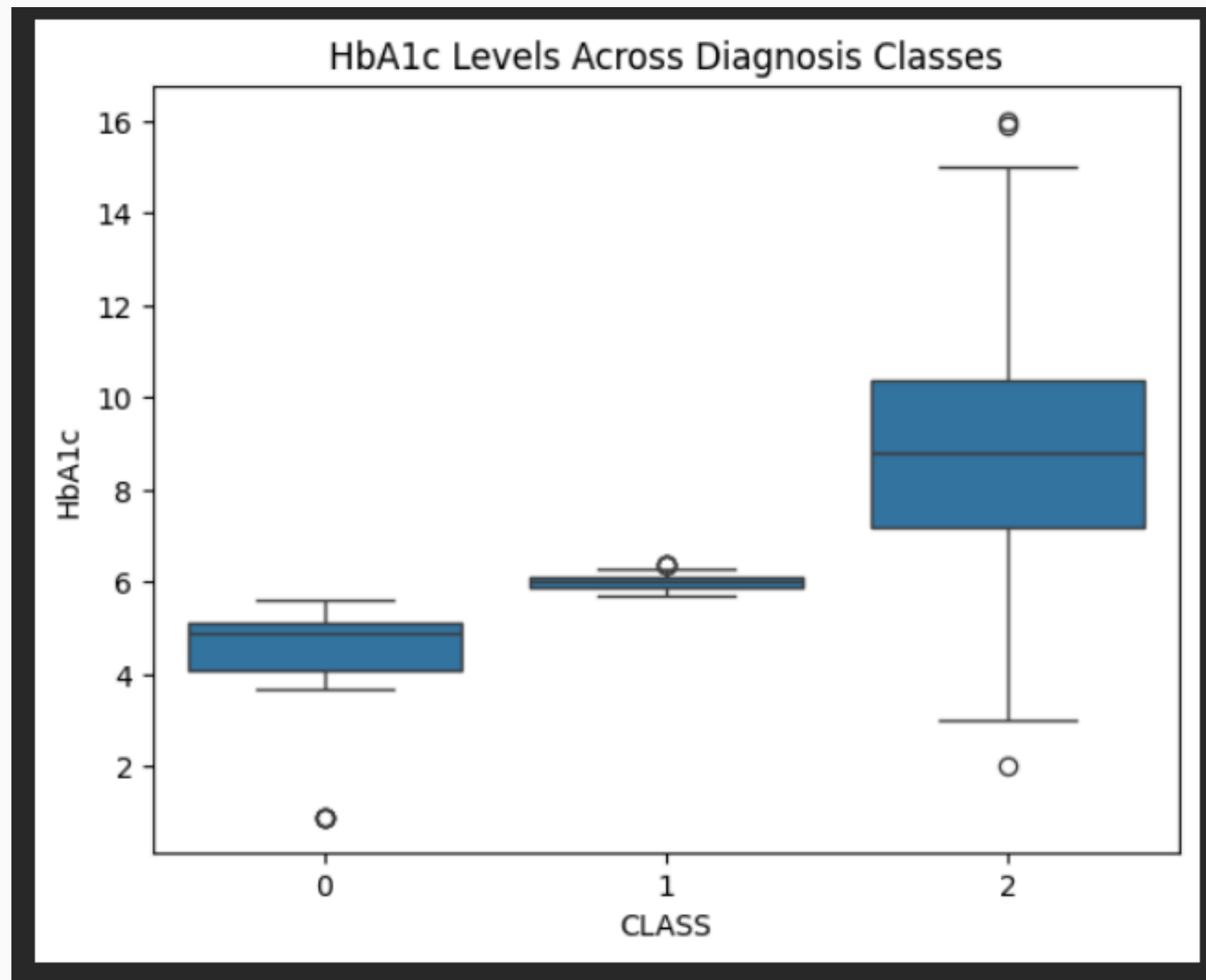
1.How many patients are in each diagnosis Class ?



- Class 0 : Non-Diabetic
- Class 1 : Pre-Diabetic
- Class 2 : Diabetic

2.DATA INSIGHTS

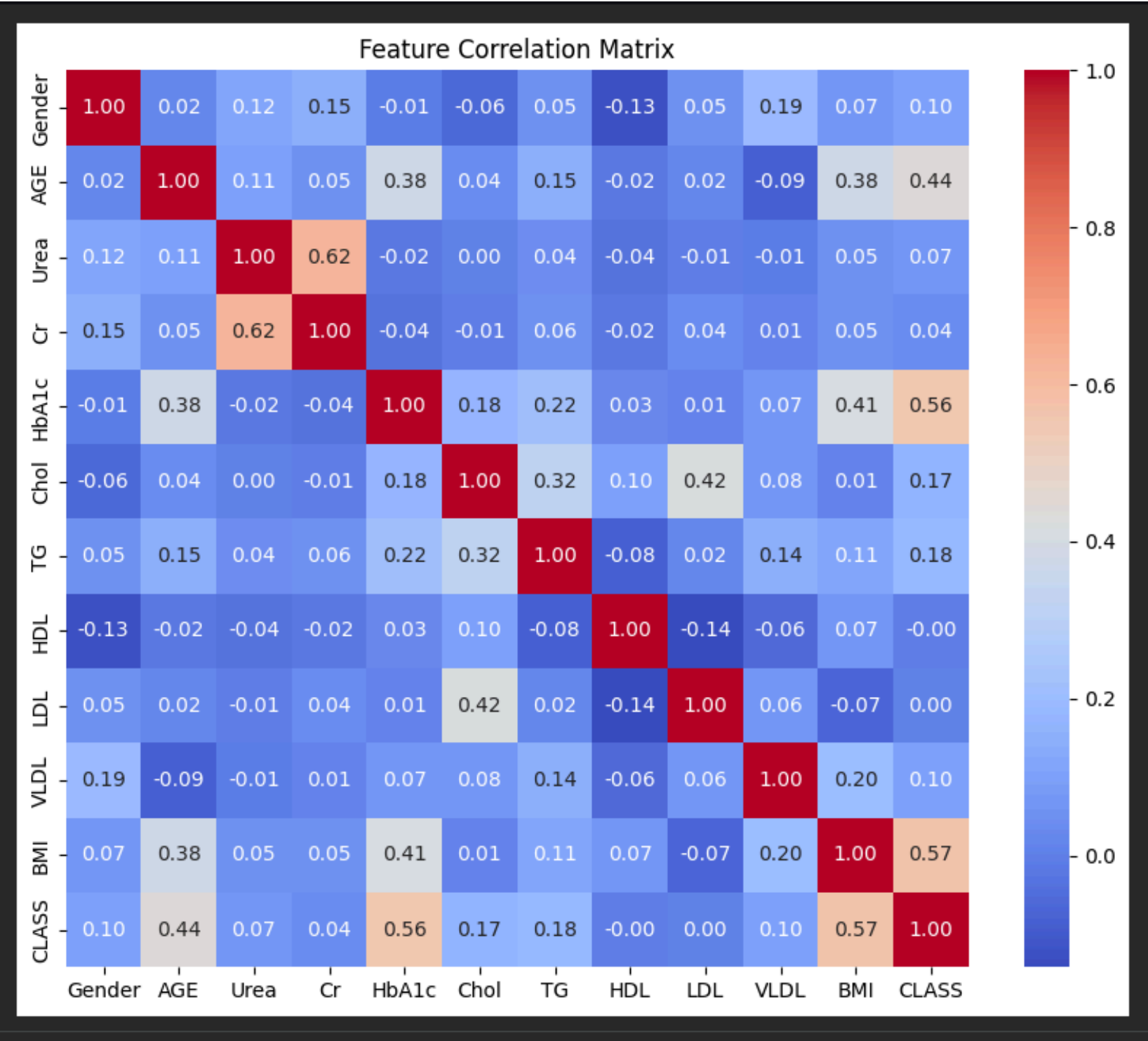
2. Which HbA1c levels Range belongs to which Class?



- Class 0 (Non-Diabetic): between 4.0 and 5.2.
- Class 1 (Prediabetic): around 5.7 to 6.3.
- Class 2 (Diabetic): between 7.0 and 10.5 (reaching up to 16).
- Takeaway: Higher HbA1c levels strongly correlate with diabetes.

2.DATA INSIGHTS

3.Which features are most strongly tied to diabetes?



1.BMI (Body Mass Index) – Correlation: 0.57

2.HbA1c (%) – Correlation: 0.56

3.AGE – Correlation: 0.44

3.METHODOLOGY

Data Cleaning

- 1.Removed unnecessary columns (ID, No_Pation)
- 2.Encoded categorical variables (Gender, CLASS)
- 3.Standardized features for Logistic Regression
- 4.Split the dataset into 80% training and 20% testing

3.METHODOLOGY

Machine Learning Models

Logistic Regression:

- Used as a baseline classification model.
- Suitable for binary classification.
- Hyperparameters tuned using GridSearchCV.

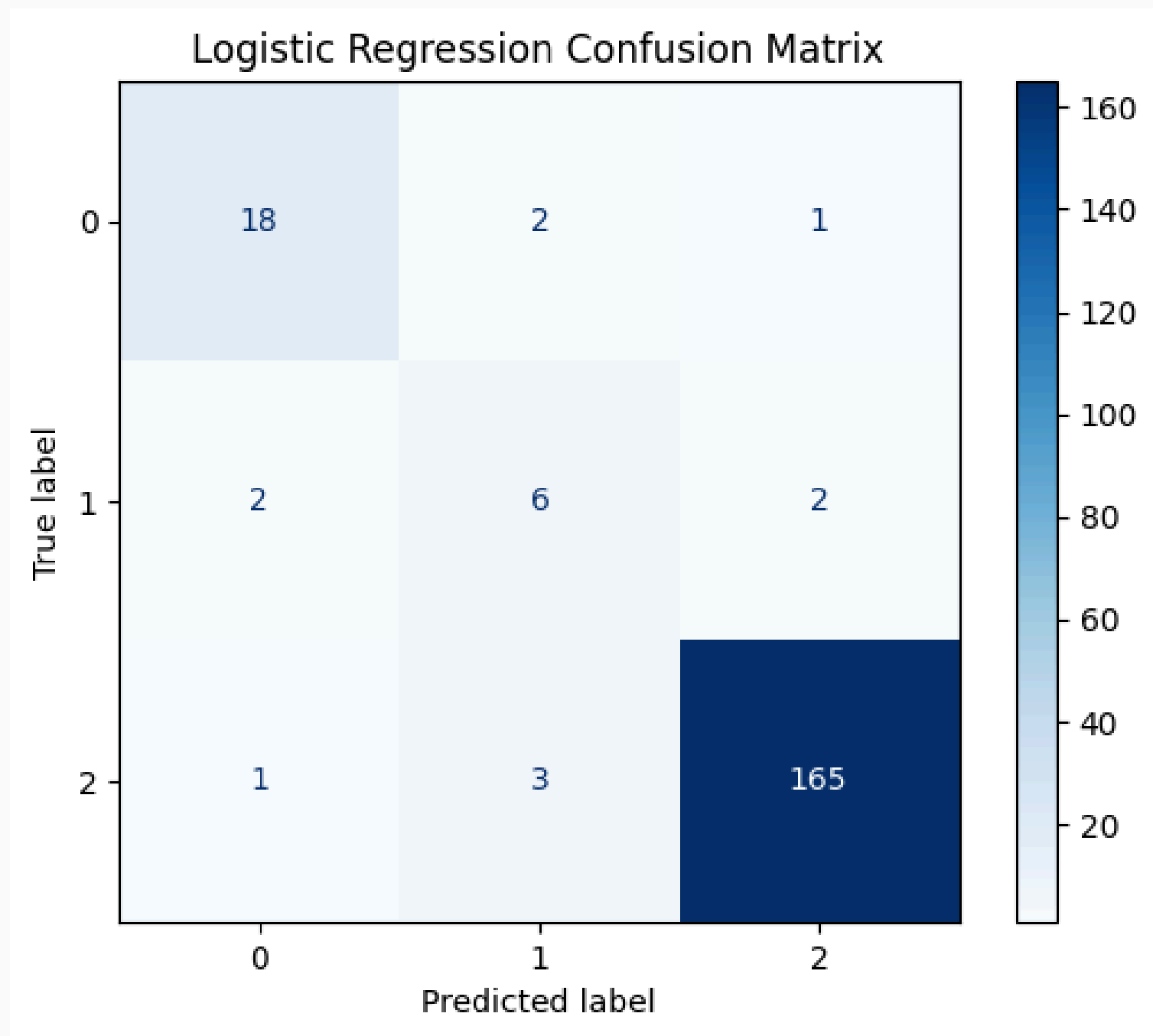
3.METHODOLOGY

- Logistic Regression:

--- Optimized Logistic Regression Performance ---					
Accuracy Score: 94.50%					
	precision	recall	f1-score	support	
0	0.86	0.86	0.86	21	
1	0.55	0.60	0.57	10	
2	0.98	0.98	0.98	169	
accuracy			0.94	200	
macro avg	0.79	0.81	0.80	200	
weighted avg	0.95	0.94	0.95	200	

3.METHODOLOGY

- Logistic Regression:



- Class 0: 18 correctly identified (True Negatives). 3 were misclassified (2 as Pre-diabetic, 1 as Diabetic).
- Class 1: 6 correctly identified (True Positives). 4 were misclassified (2 as Non-diabetic, 2 as Diabetic).
- Class 2 : 165 correctly identified (True Positives). 4 were misclassified (1 as Non-diabetic, 3 as Pre-diabetic).

3.METHODOLOGY

Machine Learning Models

Random Forest:

- Ensemble learning algorithm based on multiple decision trees.
- Handles nonlinear relationships effectively.
- Hyperparameters optimized using GridSearchCV.

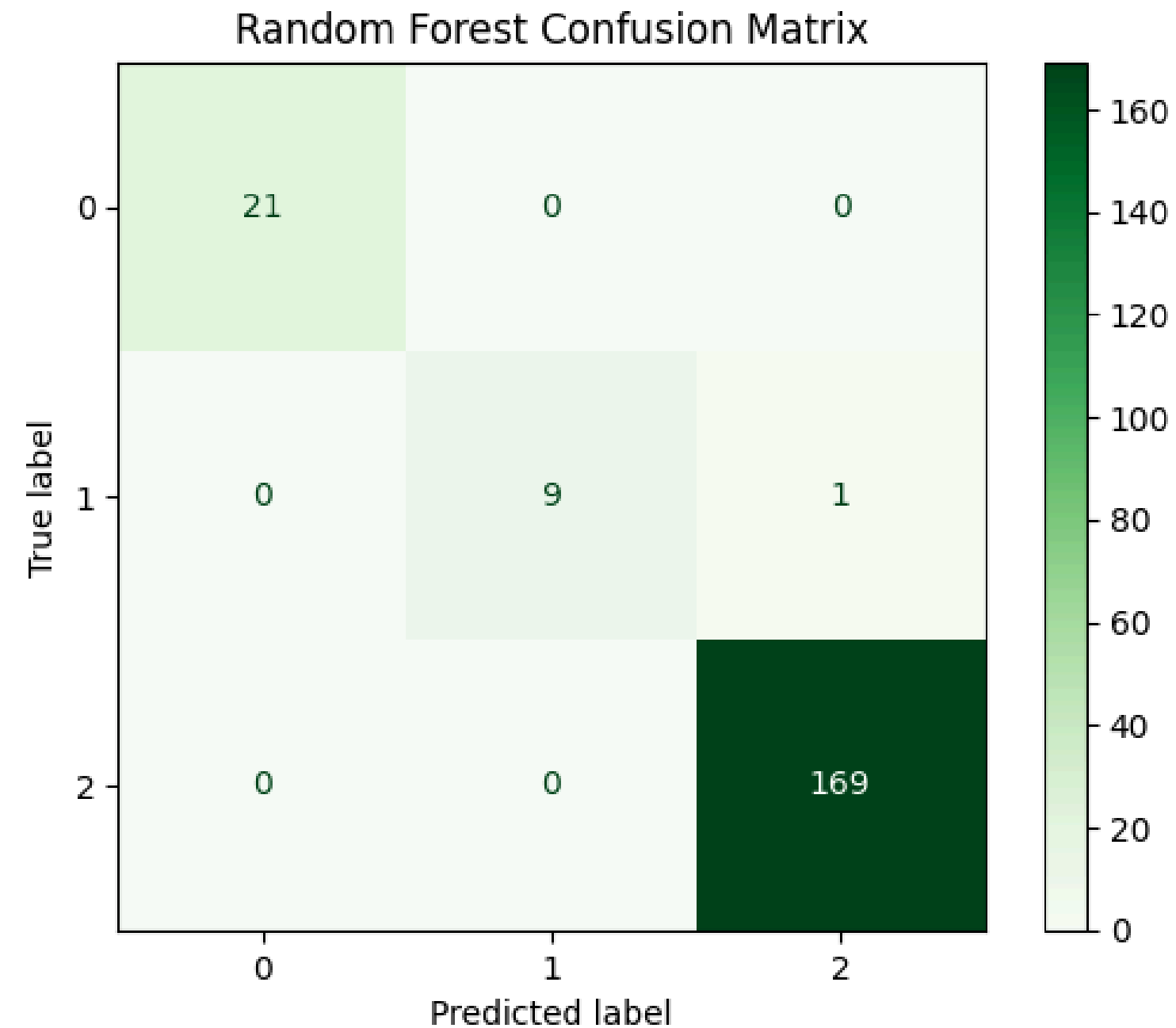
3.METHODOLOGY

- Random Forest

*** --- Optimized Random Forest Performance ---					
Accuracy Score: 99.50%					
	precision	recall	f1-score	support	
0	1.00	1.00	1.00	21	
1	1.00	0.90	0.95	10	
2	0.99	1.00	1.00	169	
accuracy			0.99	200	
macro avg	1.00	0.97	0.98	200	
weighted avg	1.00	0.99	0.99	200	

3.METHODOLOGY

- Random Forest



- Class 0 : correctly predicted all 21 of class 0.
No misclassifications for this class.
- Class 1: correctly predicted 9 of class 1. One of class 1 was misclassified as class 2 (Diabetic).
- Class 2 : correctly predicted all 169 of class 2.
There were no misclassifications for this class.

4.COMPARATIVE ANALYSIS

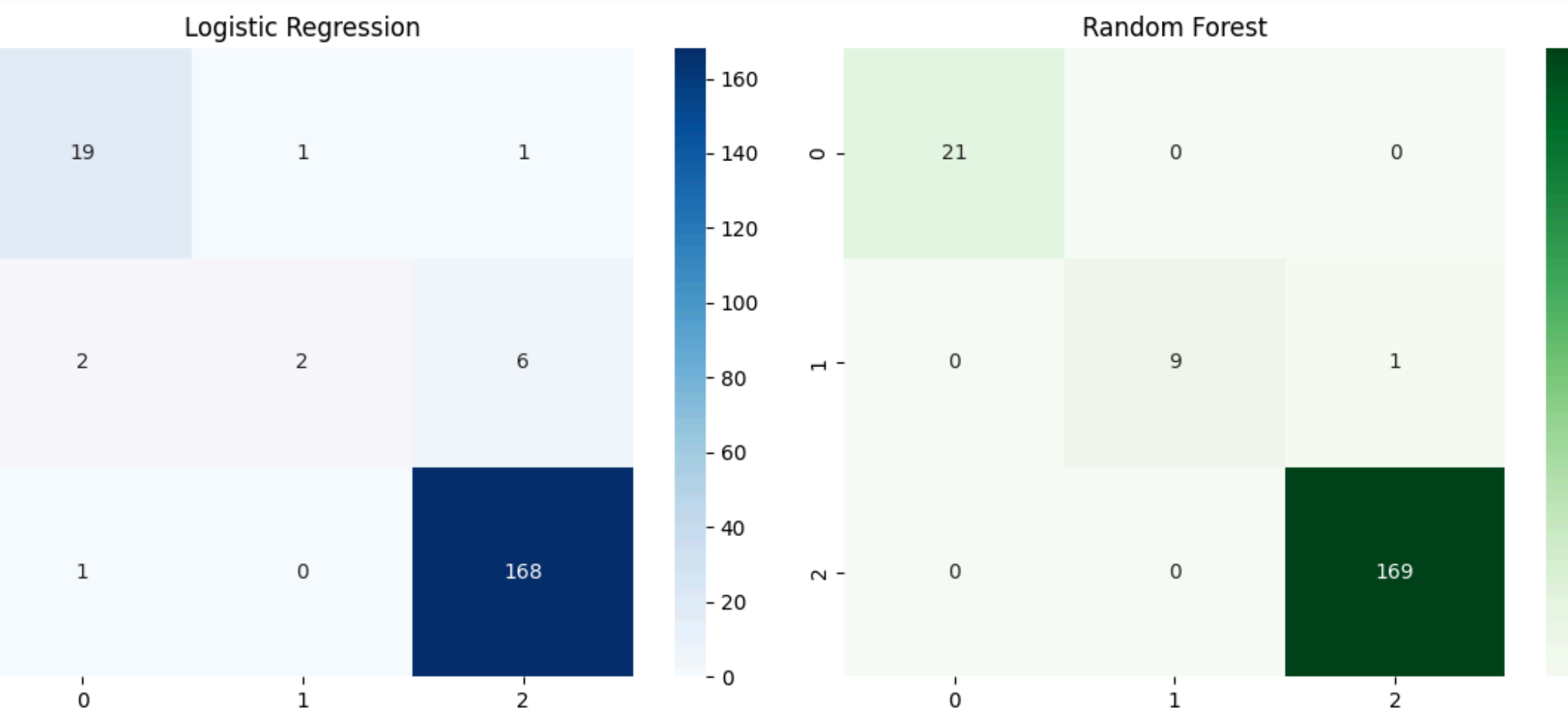
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Accuracy Score: 99.50%

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4.COMPARATIVE ANALYSIS



Evaluation	Logistic Regression	Random Forest
Class 0	19/21 correct	21/21 correct
Class 1	2/10 correct	9/10 correct
Class 2	168/169 correct	169/169 correct
Misclassifications	11	1
Overall Performance	Good	Excellent

4.COMPARATIVE ANALYSIS

Evaluation Metric	Logistic Regression	Random Forest
Overall Accuracy	94.50%	99.50%
Correct Predictions	189	199
Misclassifications	11	1
Best F1-Score (Class 0)	0.86	1
Best F1-Score (Class 1)	0.57	0.95
Best F1-Score (Class 2)	0.98	1
Overall Best Model	—	Random Forest



**THANK YOU
VERY MUCH**